CONFAI 2026 · PLAKSHA UNIVERSITY · TRACK 06 — RESPONSIBLE & ETHICAL AI

Refuse to Hallucinate

A deterministic validity firewall for Indian official survey statistics

> Every question gets a **cited verdict** — frozen with SHA-256 **before** any document is read. 1,708 verified rows · $0.00 per query · no API key, no GPU.

MoSPI's tools help you **find** the number. This firewall checks whether you are **allowed to say** it — and stops you, with a cited rule, when you are not.

QR scan for the **public demo**

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Poster + live demo · Live demo → QR · offline · no API key · no GPU · $0.00 / query

Service · MCP tool · Agent · Library

1 The problem, and the fix

Retrieval over official PDFs is solved. **Governance is not.** Unconstrained RAG over sovereign statistics fails in three ways — observed live in MoSPI's StatsDoc pilot (*archived transcripts, chats 2472–2480*):

- ✕ **Inconsistent figures in one answer.** Measured MP MPCE quoted as both 3,158 and 5,589 for urban, and 5,538 vs 4,987 for "All", in a single response (chat 2474).
- ✕ **Unguided cross-round arithmetic.** ~40% MPCE growth quoted across the HCES 2011-12 → 2023-24 break (paper → 3-visit CAPI) with no caveat.
- ✕ **Retrieval poisoning.** A poisoned document chunk can override guardrails and flip a verdict.

The thesis. You cannot fix a probabilistic system with more probabilistic prompts. The decision — *is this claim allowed?* — must come from **versioned, deterministic rules**, before generation. Models translate; rules judge.

The fix, end to end — a real response from the running service

"mpce for madhya pradesh"

VERDICT **PASS_WITH_NUMBER** RULE R_OK

3,522 Rs/month

Madhya Pradesh · rural · with imputation. The slice is not narrowed — the open question returns **every published cell**: 3,522 / 5,589 with imputation, 3,441 / 5,538 without — **4 cells, 4 row ids**.

```
source HCES 2023-24 factsheet, Statement 7
row HCES_2023_24_MPCE_2023-24_rural_with_MadhyaPradesh
verdict hash 7061f3ac… · tier VERIFIED
```

No number is shown without a row id, a PDF page and a hash.

EVERY REFUSAL IS CITED

A verdict is never bare prose — it carries a stable reason code.

- `R_OK` — verified row matched
- `R_NO_CURATED_ROW` — not curated → coverage note, no figure
- `R_NUMERIC_MISMATCH` — asserted number ≠ the store
- `R_COMPARABILITY_*` — survey break → mandatory caveat
- `R_SCOPE_OUT / R_FORECAST` — forecast, causal, political → refused
- `R_PROMPT_INJECTION` — injection refused before any lookup
- `R_METADATA_*` — taxonomy check failed
- `R_LOW_EXTRACTION_CONF` — extraction confidence too low

2 The firewall

1 **Question** — natural language, any register

↓

2 **Typed claim** — translator: models translate, never judge

↓

3 · Firewall — five stages, fixed order, first rule fires

- **Safety-A** — injection · EN / Hindi / Hinglish
- **Scope** — forecast · causal · politics
- **Comparability** — survey breaks
- **Metadata** — taxonomy check
- **Numeric** — verified rows only

↓

4 · SHA-256 freeze — computed **BEFORE** retrieval

```
hash = SHA-256(claim + decision + reason + rules)
```

↓

5 Retrieve (untrusted document text)

↓

6 **Templated render + citations** — WARN/REFUSE wording is fixed

Safety-B · freeze before retrieval + figure attestation

The verdict hash is locked before any document is read. A planted "ignore the caveat" chunk changes nothing — the hash is byte-identical with or without poisoning. A second hash binds the **number you see** to its cell.

```
frozen_hash = SHA-256(claim ⊕ decision ⊕ sub_code ⊕ reason ⊕ rule_version)
```

```
attestation = SHA-256(frozen_hash ⊕ row_id ⊕ value ⊕ unit ⊕ pdf_sha256)
```

Field-guarded cache · no safe threshold

Danger-pair and paraphrase cosine ranges overlap on both measured embedders, so **no similarity threshold is safe**. Reuse only on exact critical-field match.

```
0 collisions across 19 danger pairs · 1708 verified rows (55 touched by the golden set)
```

Generalised, not hard-coded

The translator resolves **every** state/UT the store carries, in any spelling — "mp", "up", "m.p.", "orissa", "maharastra" — and answers one state or several at once. An open slice returns **every published cell**, never one narrowed figure.

```
"MPCE in Madhya Pradesh, UP, Maharashtra" → 12 cells · 12 row ids
```

3 Results

0.000 hallucination rate — 0 of 36 **must-abstain goldens** answered wrongly	**1.000** abstention recall — 36 of 36 must-abstain goldens refused or warned
100/100 verdict accuracy [Wilson 0.963–1.000]	**60**/60 numeric exact match, on the 60 numeric goldens
0/19 danger-pair cache collisions	**1708**/1708 verified rows with provenance
1.000 Safety-A P/R/F1 · n=28 (EN · hi · hi-en)	**$0.00** marginal cost · p50 < 200 ms, CPU-only

VIDHI VS STATSDOC — HONEST COMPARISON

CHECK	STATSDOC (ARCHIVED)	VIDHI (MEASURED)
Definition	"Monthly…" — matched (by chance)	statutory, pinned by rule
Consistency	3,158 vs 5,589 urban	one row id per figure
Cross-round math	~40% growth, no caveat	halts · R_COMPARABILITY_HCES_METHOD
Poisoned context	not tested — *no such row*	SHA-256 frozen pre-retrieval (tested)
Cost / latency	~$0.02–0.05 · 3.5–8 s *(est.)*	$0.00 · p50 < 200 ms

*StatsDoc rows are transcribed from archived live transcripts (chats 2472–2480, same nine prompts); Vidhi rows are measured offline. One prompt set, one session — an observation, not a benchmark. The poisoned-context row was **not** tested live.*

HONEST LIMITS

- **Coverage is the limit, not logic** — 1,708 verified rows: the PLFS Highlights battery (11 rounds × usual + CWS × LFPR/WPR/UR × 9 sector-sex cells), the TUS 2024 battery (9 ICATUS divisions), and the HCES 2023-24 state + trend columns. State-wise MPCE covers **all 36 states/UTs** in any spelling. Uncovered slices return `WARN_NO_DATA` + a coverage note, never a guess.
- Heuristic extraction breaks on exotic phrasing; an LLM translator port exists (`VIDHI_USE_LLM=1`), unmeasured.
- No load test yet; the eSankhyiki live-API path inherits third-party cert-verification weakness (query path is offline).

First rule fires — two real traces from the running service

"Compare average MPCE in 2011-12 vs 2023-24."

1 Safety-A	pass — no injection signal
2 Scope	pass — descriptive
3 Comparability	**WARN** R_COMPARABILITY_HCES_METHOD — 68th-round paper vs 3-visit CAPI

"Will unemployment fall in 2027?"

1 Safety-A	pass — no injection signal
2 Scope	**REFUSE** R_FORECAST — no such round exists

A stage that fires **ends** the adjudication — stages 4 and 5 are never reached in either trace. The order *is* the guarantee: a later stage can never rescue a claim an earlier rule refused.

Six guarantees

1. Rules judge, models translate
2. Freeze before retrieval
3. No cosine threshold is safe
4. No PASS without provenance
5. Ambiguity → every cell, never one guess
6. Honest coverage — never a guess

REPRODUCE EVERYTHING (OFFLINE)

```
# no API key · no GPU · no network
pip install -r requirements.txt
python -m pytest tests/ -q        # 117 passed · every store row reachable
python vidhi/eval.py               # 100 golden + 19 danger + 28 red-team + Safety-B
python -m uvicorn vidhi.service:app --port 8001  # → /demo
```

TRY THE LIVE DEMO

Live demo — scan the QR (or `localhost:8001/demo` on the laptop): ask a PLFS / HCES / TUS / MIS question. Watch the verdict freeze, the citations resolve, and every figure name its source row.

MCP: `python -m vidhi.mcp_server` (4 governance tools) · **Agent:** `POST /analyze` for a policy briefing.

Repo: `github.com/ajayc-iima/msai` — 117 tests, 1,708 sha256-bound rows, the four source PDFs committed.

THE QUESTION THIS WORK WON'T HALLUCINATE

Ask the hard question — expect `WARN_NO_DATA` + a coverage note, never a guess. Inject in English, Hindi or Hinglish and watch Safety-A refuse with a cited rule.

Questions and red-team attempts welcome.